Grid Facilities Procedure			
1803 Local Planning Critieria			
Distribution: PUBLIC		Document No.: **1803-3075**	**1803 Electric Cooperative, Inc.**
Version No.: 1	Version Date: 12/31/2024	Review Cycle: As Needed	Effective Date: 12/31/2024

Table of Contents

Purpose **2**
1. Responsibilities **2**
2. Scope/Applicability **3**
2.1 Background and Facility Information 3
2.2 Scope/Applicability 4
3. 1803 Voltage Criteria (Pursuant to TPL-001-5 Requirement R5) **4**
3.1 Steady State Voltage Limits and Voltage Deviations 4
3.2 Transient Voltage: Generator Low Voltage Ride-Through Capability 4
3.3 Transient Voltage: Generator High Voltage Ride-Through Capability 5
3.4 Transient Voltage: Load Low Voltage Recovery Limits 5
4. 1803 Facility Rating Methodology **5**
4.1 Facility Rating Methodology 5
4.2 Facility Rating Criteria 5
5. 1803 Stability Criteria (Pursuant to TPL-001-5 Requirement R6) **6**
5.1 Stability Criteria 6
5.2 Cascading Outage Definition Criteria 6
5.3 Voltage Instability Definition Criteria 7
5.4 Uncontrolled Islanding Definition Criteria 7
6. 1803 System Planning Criteria **7**
6.1 Steady State Reliability 7
6.2 System loads 8
6.3 Load Distance Serving Factors (LDSF) 9
6.4 Relay Loadability Criteria 9
7. 1803 Power Quality Criteria **9**
7.1 System Frequency Variation 9
7.2 Transient Voltage Fluctuation 10
7.3 Voltage Waveform Quality 11
8. Supporting Information **12**
8.1 Operationally Affected Parties 12
8.2 Training 13
8.3 References 13
8.4 Definitions 14
8.5 Retention of Evidence and Documentation 15
8.6 Version History 15
9. Periodic Review Procedure **16**
9.1 Review Criteria 16
9.2 Frequency 16
9.3 Incorporation of Changes 16
Approval **16**
Endnotes **16**

PURPOSE

1803 Electric Cooperative, Inc. (1803) is registered as a Transmission Planner (TP) within the Midcontinent Independent System Operator, Inc (MISO) Planning Coordinator Area. 1803 participates in both SERC and MISO annual model development and transmission system planning processes and utilizes models from those processes to conduct Planning Assessments.

1803 has prepared this document to outline and define the local planning criteria, under Requirements R5 and R6 of the NERC TPL-001 Reliability Standard, used for planning the 1803 Transmission System.

1. RESPONSIBILITIES

ROLE	RESPONSIBILITY
Chief Operating Officer	Provides executive oversight of the 1803 operating environment and is responsible for approving this document.
Manager of Transmission Planning	Responsible for this document and its contents, and for periodic reviews to ensure adequacy. Responsible for ensuring accurate system data is provided as requested. Responsible for publication and distribution of annual Planning Assessments.
1803 Asset Management Personnel (AMP)	Comprised of 1803 managers and staff responsible for implementation, management, and day-to-day 1803 local operations, engineering, and maintenance. Responsible for adhering to policies, procedures and documents where described. Includes personnel acting on behalf of 1803 through operating or contractual agreements.
Planning Coordinator (PC) MISO	Responsible for coordinating and integrating transmission Facilities and service plans, resource plans, and Protection Systems in Long-Term and Near-Term Transmission Planning Horizon studies.

ROLE	RESPONSIBILITY
Transmission Planner (TP) 1803	Responsible for developing long-term plans for the reliability (adequacy) of the interconnected bulk electric transmission systems within its area. 1803 and MISO have *CFR00143*, that identifies the compliance responsibilities each entity has as a TP.

2. SCOPE/APPLICABILITY

2.1 Background and Facility Information

2.1.1 1803 Electric Cooperative, Inc. (1803) is a Louisiana electric cooperative incorporated on April 1, 2019, who is currently owned by six electric cooperatives listed below:

2.1.1.1 Beauregard Electric Cooperative, Inc. (BEC);

2.1.1.2 Claiborne Electric Cooperative, Inc. (CLA);

2.1.1.3 Northeast Louisiana Power Cooperative, Inc. (NEL);

2.1.1.4 South Louisiana Electric Power Association (SLC);

2.1.1.5 Washington-St. Tammany Electric Cooperative, Inc. (WST); and

2.1.1.6 Pointe Coupee Electric, Inc. (PC).

2.1.2 1803 is a NERC registered Balancing Authority (BA), Distribution Provider (DP), Transmission Owner (TO), and TP.

2.1.3 1803 has contracted with Gridforce Energy Management, LLC (GEM) to provide NERC registered TOP services and GEM is the registered TOP. GEM also will provide the necessary Control Center designated services to perform the 1803 registered BA functions.

2.1.4 1803 has contracted with ACES Power, LLC (ACES) for power market services for energy delivered.

2.1.5 1803 is a member of MISO. MISO provides services that include the following NERC registered duties: BA, Planning Authority (PA)/PC, Reliability Coordinator (RC), Resource Planner (RP), Transmission Operator (TOP), TP, and Transmission Service Provider (TSP).

2.1.5.1 1803 and MISO have CFR00001 that outlines the compliance responsibilities between the entities for the BA function.

2.1.5.2 1803 and MISO have CFR00143 that outlines the compliance responsibilities between the entities for the TP function.

This procedure is limited to the detailed functions that are performed by and for 1803. The processes within the procedure are primarily associated with Bulk Electric System (BES).

2.2 Scope/Applicability

2.2.1 NERC reliability standards require the operation of 1803's BES facilities or MISO Transferred facilities to be within the established planning criteria guidelines of this document.

2.2.2 1803 further applies the guidelines in Section 3.1 and Section 4.2 to Non-BES or MISO Non-Transferred Facilities to ensure reliable service to its members.

3. 1803 Voltage Criteria (Pursuant to TPL-001-5 Requirement R5)

3.1 Steady State Voltage Limits and Voltage Deviations

3.1.1 Normal Low Voltage Limit (p.u.): 0.95

3.1.2 Normal High Voltage Limit (p.u.): 1.04

3.1.3 Emergency Low Voltage Limit (p.u.): 0.9

3.1.4 Emergency High Voltage Limit (p.u.): 1.05

3.1.5 Post Contingency Max. Voltage Deviation (p.u.): 0.14

3.2 Transient Voltage: Generator Low Voltage Ride-Through Capability[a]

3.2.1 0.00 to 0.15 seconds (p.u.): 0.00

3.2.2 0.15 to 0.30 seconds (p.u.): 0.45

3.2.3 0.30 to 2.00 seconds (p.u.): 0.65

3.2.4 2.00 to 3.00 seconds (p.u.): 0.75

[a] Based on Attachment 2 of the NERC PRC-024 Reliability Standard

3.2.5 Beyond 3.00 seconds (p.u.): 0.90

3.3 Transient Voltage: Generator High Voltage Ride-Through Capability[b]

3.3.1 0.00 to 0.20 seconds (p.u.): 1.20

3.3.2 0.20 to 0.50 seconds (p.u.): 1.175

3.3.3 0.50 to 1.00 seconds (p.u.): 1.15

3.3.4 Beyond 1.00 seconds (p.u.): 1.10

3.4 Transient Voltage: Load Low Voltage Recovery Limits

3.4.1 0.00 to 20.00 seconds after fault clearing (p.u.): 0.70

3.4.2 Beyond 20.00 seconds after fault clearing (p.u.): 0.90

4. 1803 Facility Rating Methodology

4.1 Facility Rating Methodology

4.1.1 1803 establishes and records assumptions and inputs used in processes to determine Facility Ratings.

4.1.2 1803 identifies the most limiting element for each Facility and shares these values with MISO and the adjacent Transmission Owners.

4.2 Facility Rating Criteria

4.2.1 1803 utilizes a Normal Rating (Rate A) in transmission planning models for identifying overloaded transmission elements during base case conditions where all transmission elements are in service.

4.2.2 1803 utilizes an Emergency Rating (Rate B) in transmission planning models for identifying overloaded transmission elements for all contingency analysis.

[b] Based on Attachment 2 of the NERC PRC-024 Reliability Standard

5. 1803 STABILITY CRITERIA (PURSUANT TO TPL-001-5 REQUIREMENT R6)

5.1 Stability Criteria

5.1.1 The existing system, following interrupting operations necessary to clear a fault, will maintain synchronism and exhibit acceptable damping for all single-line-to-ground and three-phase faults with normal clearing, and for all single-line-to-ground faults with delayed clearing, as specified in NERC standards. Planning events will not cause a Cascading Outage as defined in Section 5.2.

5.1.2 1803's criteria requires a minimum 3% dampening for all power system oscillations for P1-P7 events. The machine rotor angle damping ratio is determined by modal analysis (e.g. Prony analysis or equivalent). Alternatively, the Rotor Angle Oscillation Damping Factor or Successive Positive Peak Ratio (SPPR) can be calculated directly from the rotor angle, where the rotor angle response allows such direct calculation. For a disturbance with a fault, the SPPR must be less than 0.95 or the damping factor must be greater than 5%. For a disturbance without a fault, the SPPR must be less than 0.9 or the damping factor bus be greater than 10%.

5.2 Cascading Outage Definition Criteria

5.2.1 For contingency assessments, including those assessments identified as simulated planning events (P1 through P7) or multiple and extreme events, additional transmission system analyses may be required. These assessments are to limit the possibility of potential cascading events should such an event result following the loss of 300 MW of load (consequential or non-consequential), or voltage instability indicated by a non-convergent case.

5.2.2 1803 defines these criteria for identifying cascading outages on the BES:

5.2.2.1 Contingency events with significant system impacts (i.e., loss of 300 MW of load) are monitored for any post-contingency BES transmission facility loading that exceeds 115% of that facility's continuous rating. During these assessments, the facility with the highest transmission facility loading is simulated as opened.

5.2.2.2 Analysis of post contingency cases are performed to identify additional BES Facilities that exceed 115% of that facility's continuous rating.

5.2.2.2.1 During these analyses, the 1803 Facility with the highest transmission facility loading is simulated as opened.

5.2.2.2.2 Each analysis is concluded once all facilities are below 115% transmission facility loading.

5.2.2.2.3 Each analysis evaluates the number of opened BES Facilities and total magnitude of load shedding simulated.

5.2.2.3 Contingency events are considered cascading when one of the following occurs:

5.2.2.3.1 Three or more facilities are simulated opened due to post contingency loading more than 115% transmission facility loading.

5.2.2.3.2 The overloaded facilities occur outside the 1803 Transmission Planning Area.

5.2.2.3.3 The case solution diverged (i.e., did not solve) due to system conditions that would lead to voltage collapse.

5.3 Voltage Instability Definition Criteria

5.3.1 Voltage Stability Maximum Transfer Limit (% of transfer at nose of PV curve): 90%

5.4 Uncontrolled Islanding Definition Criteria

5.4.1 Angular Transient Stability Minimum Damping Ratio (ζ): 0.03

5.4.2 Angular Transient Stability Critical Clearing Time Margin (cycles): 1

6. 1803 System Planning Criteria

6.1 Steady State Reliability

6.1.1 The 1803 transmission system shall be planned based on the following reliability criteria:

6.1.1.1 The system must meet voltage criteria and thermal loading performance criteria when all facilities are in service as defined in Sections 3.1 and Section 4.2, respectively.

6.1.1.2 The system must meet contingency voltage criteria and contingency thermal loading performance criteria as defined in Sections 3.1 and Section 4.2 respectively following post contingency system adjustments for any of the following N-1 contingencies:

6.1.1.2.1 Transmission Line

6.1.1.2.2 Transformer

6.1.1.2.3 Generator

6.1.1.2.4 Single Intertie

6.1.1.2.5 Shunt Device

6.1.1.2.6 Double Circuit Line

6.1.2 Post contingency system adjustments to be considered include opening or closing switches, changing transformer load taps, energizing or deenergizing capacitor banks or reactor banks, and generation redispatch.

6.1.2.1 The available range of load tap changers will be modeled in accordance with that specified on the transformer name plate.

6.1.3 Shunt capacitor and reactor banks will be added to the 1803 system according to the following criteria to support and regulate the system voltage:

6.1.3.1 Capacitor and reactor banks will be connected in delta or ungrounded wye, unless otherwise determined due to location.

6.1.3.2 Capacitor and reactor bank additions will be limited to produce a steady-state voltage step change upon energizing of 3.0% or less without adjustment of transformer load tap changers.

6.1.3.3 No more than two (2) capacitor or two (2) reactor banks will be installed between interrupting devices.

6.1.3.4 Capacitor and reactor banks may be staged and/or automatically switched to control system voltage.

6.1.3.5 Capacitor banks will not produce a voltage rise under no load conditions above the maximum voltage, and reactor banks will not produce a voltage drop under peak load conditions below the minimum voltage as specified in Section 3.1.

6.1.3.6 Power factor correction up to 0.93 and above at the point of common coupling will be conserved a valid solution to voltage criteria violations with power factor correction occurring at the distribution level.

6.2 System loads

6.2.1 System loads used for internal planning will be based upon historical system load,

historical seasonal weather patterns, and normal economics; this system load will come from the most recently approved long term load forecast.

6.2.2 Loads will be modeled using historical power factor data or power factor projections provided by our members.

6.3 Load Distance Serving Factors (LDSF)

6.3.1 1803 evaluates Load Distance Serving Factors (LDSF) when conducting its Transmission System Planning Assessments.

6.3.1.1 If there are multiple substations or delivery points served by the radial transmission line, an LDSF is to be calculated for each point, including the total load being served from and through that delivery point.

6.3.1.2 When radial transmission line loading reaches an LDSF between 100 to 150 MW-miles, efforts should be included in system long-range planning to reduce the radial exposure.

6.3.2 Establishing a looped transmission path can be considered as an alternative solution to ensure that the steady-state criteria of Section 3.1 and Section 4.2 are maintained.

6.3.2.1 No more than four (4) load stations should be served between automated interrupting devices.

6.4 Relay Loadability Criteria

6.4.1 Evaluation of BES facility tripping for relay loadability will occur when tripping of BES transmission elements result in a BES element exceeding 130% of maximum continuous facility rating. Such loading conditions will be evaluated against actual relay settings.[1]

7. 1803 Power Quality Criteria

7.1 System Frequency Variation

7.1.1 The frequency of the 1803 transmission system will be nominally 60 Hz and shall be maintained within the limits of 59.95 Hz - 60.05 Hz under normal steady-state operation.

7.1.2 Under-frequency load shedding will occur in the following steps.

7.1.2.1 59.3 Hz

7.1.2.2 59.0 Hz

7.1.2.3 58.7 Hz

7.2 Transient Voltage Fluctuation

7.2.1 Transient voltage fluctuations at a point of common coupling with a fluctuating load directly connected to the system shall not exceed the following:

7.2.1.1 Transient voltage fluctuations consisting of step changes (including motor starts) that occur frequently, shall not exceed the following voltage excursions at the point of common coupling:

- 69kV: 2.0%
- 115kV: 1.5%
- 161kV and above: 1.0%

7.2.1.2 Transient voltage fluctuations consisting of step changes (including motor starts) that occur infrequently (no more than two in any consecutive 24-hour period), shall not exceed the following excursions at the point of common coupling:

- 69kV: 3.5%
- 115kV: 2.5%
- 161kV and above: 1.5%

7.2.2 Voltage Flicker

7.2.2.1 Voltage flicker is systematic variations of the voltage envelope or a series of random voltage changes, where the magnitude does not normally exceed the steady-state voltage ranges. Voltage flicker negatively impacts lighting intensity.

7.2.2.2 A measure of the visual severity of flicker shall be determine from measurements conducted at a consumer's premises and who has complained of flicker or equivalent over a 10-minute period. The measured results shall be compared against the GE Flicker Curve cited in IEEE 1453 to determine whether the measurements exceed the "Border Lines of Irritation."

7.2.3 Voltage Recovery Criteria

7.2.3.1 For the NERC planning events P1 through P7, all BES buses shall recover above 0.9 p.u. voltage within 10 seconds after the fault clearing.

7.2.4 Voltage Dip Criteria

7.2.4.1 For the NERC planning events P1 through P7, after 0.8 p.u. voltage recovery following fault clearing, the secondary voltage dip for all BES buses shall:

- Recover to above 0.7 p.u. within 30 cycles
- Recover to above 0.8 p.u. within 2 seconds

7.2.5 Cascading criteria

7.2.5.1 The result of fault analysis of the extreme events may be considered cascading if more than six (6) BES substations cannot recover voltage above 0.9 p.u. 10 seconds after the fault clearing.

7.3 Voltage Waveform Quality

7.3.1 All plant and apparatus connected to 1803 system should be capable of withstanding the following distortions of the voltage waveform with respect to harmonic content and phase unbalance. The maximum total level of harmonic distortion on the system from all sources under both planned outage and fault outage conditions shall not exceed the values listed in Table 1 of IEEE 519:

7.3.1.1 Individual voltage distortion (%)

- <= 69kV: 3%
- > 69kV & < 161kV: 1.5%
- > 161kV: 1%

7.3.1.2 Total Voltage Distortion THD (%)

- <= 69kV: 5%
- > 69kV & < 161kV: 2.5%
- > 161kV: 1.5%

7.3.1.3 Under planned outage conditions:

7.3.1.3.1 The maximum negative phase sequence component of the phase voltage on the interconnected system should remain below 1% unless abnormal conditions prevail. Infrequent short duration peaks with a maximum value of 2% are permitted for phase imbalance, subject to prior agreement of 1803.

7.3.1.4 Under normal operating conditions:

7.3.1.4.1 The generator will withstand continuous negative phase sequence component as defined in ANSI standard C50.13 American National Standards Institute Requirements for Cylindrical Rotor Synchronous Generations

7.3.2 Current Waveform Quality

7.3.2.1 All plant and apparatus connected to the 1803 system should not produce harmonic currents in excess of the voltages shown in Tables 2, 3, and 4 of IEEE 519.

7.3.2.2 Any harmonic mitigation measures implement to meet the current distortion limits shall be sole responsibly of the interconnection facility. Any reaction power compensation devise utilized to mitigate harmonics should only be installed upon authorization from 1803.

7.3.3 Field Measurements

7.3.3.1 Effort is taken to limit the level of voltage and current harmonics in 1803's system. After facilities are in operation, problems can arise that were not present in the planning and design phase. 1803 may periodically verify adherence to the criteria using field measurements taken at a 5-minute interval or less over a minimum of 24 hours. The measurements will be used to conduct a statistical analysis against the criteria.

8. SUPPORTING INFORMATION

8.1 Operationally Affected Parties

8.1.1 1803 as the BA, DP, TO, and TP.

8.1.1.1 MISO as the BA for functions outlined in *CFR00001*.

8.1.1.2 MISO as the TP for functions outlined in *CFR00143*.

8.1.2 GEM as the TOP.

8.1.3 MISO as the PC and TP.

8.1.4 SERC Reliability Corporation (SERC) as the Regional Entity.

8.2 Training

8.2.1 There is no required training for this document.

8.3 References

Pertinent documents related to or referenced in this document.

DOCUMENT	DESCRIPTION
CFR00143	Coordinated Function Registration Agreement outlining Transmission Planner responsibilities between MISO and 1803.
NERC Glossary of Terms	As attached to the Reliability Standards, complete set. Terms extracted from the NERC Glossary that are referenced within this document have been capitalized throughout.
MISO Planning Modeling Manual: Reliability Data Requirements & Reporting Procedures	Pursuant to requirement R1.3 of MOD-032-1, this Requirements and Reporting Procedures manual is posted on the MISO website at the following location: https://www.misoenergy.org/planning/planning-modeling/
Operating Protocols of the CFR Agreement Transmission Planners and MISO	Document developed by MISO and its TPs that summarizes the assignment of individual Transmission Planner compliance responsibilities between entities.

DOCUMENT	DESCRIPTION
NERC Dynamic Modeling Recommendations	Document historically used to establish requirements and criteria for the creation of Interconnection-wide base cases by MOD-032 designees. Appendix A of this document has the list of dynamic models deemed unacceptable following varying observations.
Appendix K of the MISO Transmission Planning Business Practices Manual	Appendix within MISO BPM-020 for defining the MISO Default Planning Criteria for voltage and Cascading characteristics.
MISO MOD Access Request Form	Documents completed by requesting personnel who seek access to the MISO MOD Software Portal. Appropriate form is selected based on use, either for reliability purposes or as a Market Participant.
IEEE 519-2022	Standard for Harmonic Control in Electric Power Systems
IEEE 1453-2022	Standard for Measurement and Limits of Voltage Fluctuations and Associated Light Flicker on AC Power Systems

8.4 Definitions

Acronyms or definitions specific to this procedure that are not generally used globally in 1803 operating processes:

TERM	DEFINITION
Load Distance Serving Factors (LDSF)	Arithmetical product of a radial transmission line length (in miles) times the last peak load experienced on the line (in MW). The LDSF is recorded in MW-miles and is a numerical value

TERM	DEFINITION
	used to assess the criticalness of electric load to the length of the transmission line.
Planning Assessment	Documented evaluation of future Transmission System performance and Corrective Action Plans to remedy identified deficiencies.
MISO Transferred Facility	BES Facilities transferred to MISO's functional control pursuant to Appendix H (Transmission System Facilities) of the Transmission Owners Agreement Transferred Transmission Facilities
MISO Non-transferred Facility	Non-BES Facilities that are not transferred to MISO's functional control pursuant subject to Appendix G (Agency Agreement) of the Transmission Owners Agreement Transferred Transmission Facilities

8.5 Retention of Evidence and Documentation

8.5.1 The following evidence records shall be retained related to this document through the last regulatory audit cycle:

8.5.1.1 Current and prior versions of this document, as retained by the 1803 AMP.

8.5.1.2 Current and prior versions of the Reference documents, as retained by the 1803 AMP.

8.5.1.3 Records associated with this document's implementation.

8.6 Version History

Version #	Change	By	Date
1	Document origination	PCS/1803	12/31/2024

Version #	Change	By	Date

9. Periodic Review Procedure

9.1 Review Criteria

9.1.1 This criteria shall be reviewed as needed to assure adherence to the current enforceable NERC Reliability Standards listed in the Endnotes Section or any changes to the documentation listed in the References Section. Any modifications will be supplied to MISO in accordance with the MISO Tariff.

9.2 Frequency

9.2.1 The 1803 Local Planning Criteria shall be reviewed and updated as needed.

9.3 Incorporation of Changes

9.3.1 The plan will be effective as indicated in the document's effective date heading section area.

Approval

Approved By	Signature	Date
Ron Repsher	**Signature:** *Ron Repsher*	12/31/2024
	Email: ron.repsher@1803electric.coop	

Endnotes

[1] TPL-001-5.1, R3.3

1803-3075 - 1803 Local Planning Criteria - Execution Version 12312024

Final Audit Report 2024-12-31

Created:	2024-12-31
By:	Matt Tillman (matt.tillman@1803electric.coop)
Status:	Signed
Transaction ID:	CBJCHBCAABAAjqdP5gGBHFD6uO6BQhveHe8q4d_ZOWuZ

"1803-3075 - 1803 Local Planning Criteria - Execution Version 1 2312024" History

Document created by Matt Tillman (matt.tillman@1803electric.coop)
2024-12-31 - 10:36:31 PM GMT

Document emailed to Ron Repsher (ron.repsher@1803electric.coop) for signature
2024-12-31 - 10:36:35 PM GMT

Email viewed by Ron Repsher (ron.repsher@1803electric.coop)
2024-12-31 - 10:52:04 PM GMT

Document e-signed by Ron Repsher (ron.repsher@1803electric.coop)
Signature Date: 2024-12-31 - 10:52:21 PM GMT - Time Source: server

Agreement completed.
2024-12-31 - 10:52:21 PM GMT

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